Accenture Technical Mock Test — Set 6

45 Questions • Computer Science • Networking • Pseudo Code

Pattern aligned to the supplied Accenture mock materials. Questions are uniquely authored across the 10 sets.

Section 1 — Computer Science

Q1. What is the result of Java expression `"A".repeat(5).length()`?

A. 5
B. 6
C. 10
D. Compilation error

Q2. Which SQL operator tests whether a value belongs to a list?

A. EXISTS ONLY
B. BETWEEN ONLY
C. IN
D. LIKE

Q3. Which container technology packages an application with its dependencies into an image?

A. Git
B. Jenkins
C. Maven
D. Docker

Q4. `Car.start()` provides a subclass-specific implementation of `Vehicle.start()`. Which feature is shown?

A. Method overloading
B. Aggregation
C. Constructor chaining
D. Method overriding

Q5. Which interface is commonly used to sort objects using a natural ordering?

A. Serializable
B. Runnable
C. Cloneable
D. Comparable

Q6. Which index type is primarily designed for equality/range lookups in many relational databases?

A. B-tree
B. Heap file
C. Stack index
D. Bitmap only

Q7. Which join preserves every row from the right table while matching records from the left where possible?

A. INNER JOIN
B. LEFT JOIN
C. RIGHT JOIN
D. CROSS JOIN

Q8. Which primitive type stores a 64-bit floating-point value?

A. decimal
B. long
C. double
D. float

Q9. Methods named `send(String)` and `send(String,int)` coexist in a Java class. What is this called?

A. Method overloading
B. Method overriding
C. Encapsulation
D. Composition

Q10. Which join would you use to keep every row from `Departments` while matching employees when available?

A. INNER JOIN
B. LEFT JOIN
C. CROSS JOIN
D. FULL OUTER JOIN

Q11. Which HTTP status code indicates a successful creation of a resource?

A. 204
B. 201
C. 302
D. 200

Q12. Which status code indicates server-side internal failure?

A. 401
B. 400
C. 204
D. 500

Q13. Which SQL clause filters rows before grouping?

A. WHERE
B. GROUP BY
C. HAVING
D. ORDER BY

Q14. Which collision-resolution technique stores multiple hash-table entries in linked lists/buckets?

A. Chaining
B. Partitioning
C. Indexing
D. Paging

Section 2 — Fundamentals of Networking

Q15. Which AAA component records user/network activity for auditing?

A. Authorization
B. Accounting
C. Authentication
D. Addressing

Q16. Which routing protocol is commonly used to exchange routes between autonomous systems?

A. RIP
B. BGP
C. OSPF
D. EIGRP only

Q17. Which EAP method uses client and server certificates for mutual authentication?

A. CHAP
B. EAP-TLS
C. PAP
D. PEAP

Q18. Which protocol securely accesses a remote shell?

A. FTP
B. SSH
C. Telnet
D. SNMP

Q19. Which protocol commonly uses TCP 23 and is insecure for remote login?

A. RDP
B. SSH
C. Telnet
D. SFTP

Q20. Which protocol commonly uses UDP 161 for queries?

A. TFTP
B. DHCP
C. DNS
D. SNMP

Q21. Which system detects suspicious traffic and can actively block it inline?

A. DNS server
B. IDS
C. Hub
D. IPS

Q22. Which TCP flag is normally used to initiate a connection?

A. ACK
B. FIN
C. SYN
D. RST

Q23. Which technology filters HTTP requests based on application-layer rules?

A. Layer-2 switch
B. NTP server
C. Web Application Firewall
D. DHCP relay

Q24. What is the broadcast address of 10.0.0.128/27?

A. 10.0.0.255
B. 10.0.0.159
C. 10.0.0.127
D. 10.0.0.160

Q25. At which OSI layer do TCP and UDP operate?

A. Data Link layer
B. Transport layer
C. Network layer
D. Application layer

Q26. Which service resolves a hostname using a local cache before querying upstream DNS?

A. NTP client
B. RADIUS
C. DHCP relay
D. Recursive DNS resolver

Q27. Which topology provides redundant paths for high availability?

A. Mesh
B. Point-to-point only

C. Single-star

D. Bus

Q28. Which mechanism controls the sender based on the receiver's available buffer?

A. NAT

B. Routing

C. Flow control

D. ARP

Section 3 — Pseudo Code

Q29. What value is printed?

```
count ← 0
FOR i ← 1 TO 14
IF i MOD 2 = 0
count ← count + 1
END IF
END FOR
PRINT count
```

A. 14

B. 8

C. 6

D. 7

Q30. What is printed after the assignments?

```
x ← 4
y ← 7
x ← y
y ← 2
PRINT x + y
```

A. 6

B. 9

C. 5

D. 11

Q31. What is printed?

```
x ← 29
PRINT x MOD 6
```

A. 6

B. 6

C. 5

D. 4

Q32. What position is printed for the target?

```
A ← [7, 12, 3, 6]
target ← 12
FOR i ← 0 TO 3
IF A[i] = target
PRINT i + 1
BREAK
END IF
END FOR
```

A. 2

B. 3

C. 0

D. 1

Q33. What is printed for the string `REFER`?

```
s ← "REFER"
r ← reverse(s)
IF s = r
PRINT "YES"
```

```
ELSE
PRINT "NO"
END IF
```

A. EMPTY

B. YES

C. ERROR

D. NO

Q34. What value is printed?

```
a ← 15
b ← 9
IF a > b
PRINT a
ELSE
PRINT b
END IF
```

A. 9

B. 24

C. 15

D. 6

Q35. What is printed?

```
n ← 6
fact ← 1
FOR i ← 1 TO n
fact ← fact * i
END FOR
PRINT fact
```

A. 721

B. 720

C. 36

D. 714

Q36. What value is printed?

```
A ← [12, 5, 8, 2]
mx ← A[0]
FOR i ← 1 TO 3
IF A[i] > mx
mx ← A[i]
END IF
END FOR
PRINT mx
```

A. 4

B. 2

C. 27

D. 12

Q37. What is the final value of sum?

```
sum ← 0
FOR i ← 0 TO 12 STEP 2
sum ← sum + i
END FOR
PRINT sum
```

A. 7

B. 12

C. 44

D. 42

Q38. How many times is the inner statement executed?

```
count ← 0
FOR i ← 1 TO 3
FOR j ← 1 TO 5
count ← count + 1
END FOR
```

```
END FOR
PRINT count
```

A. 5

B. 14

C. 15

D. 8

Q39. What is printed by the recursive function for n = 8?

```
FUNCTION F(n)
IF n <= 1
RETURN n
RETURN F(n-1) + F(n-2)
END FUNCTION
PRINT F(8)
```

A. 21

B. 8

C. 20

D. 22

Q40. What value is printed?

```
A ← [3, 9, 4, 7]
mx ← A[0]
FOR i ← 1 TO 3
IF A[i] > mx
mx ← A[i]
END IF
END FOR
PRINT mx
```

A. 23

B. 3

C. 9

D. 4

Q41. How many vowels are counted in `JAVA`?

```
s ← "JAVA"
count ← 0
FOR each ch IN s
IF ch is a vowel
count ← count + 1
END IF
END FOR
PRINT count
```

A. 2

B. 0

C. 3

D. 4

Q42. What is the final printed value?

```
n ← 5678
sum ← 0
WHILE n > 0
digit ← n MOD 10
sum ← sum + digit
n ← n DIV 10
END WHILE
PRINT sum
```

A. 26

B. 78

C. 4

D. 27

Q43. Using bitwise XOR, what is printed?

```
x ← 10
y ← 6
```

```
PRINT x ^ y
```

A. 4

B. 12

C. 16

D. 60

Q44. What is printed by the pseudocode?

```
x ← 20
sum ← 0
FOR i ← 1 TO x
sum ← sum + i
END FOR
PRINT sum
```

A. 400

B. 211

C. 210

D. 209

Q45. What value is printed?

```
n ← 9001
rev ← 0
WHILE n > 0
d ← n MOD 10
rev ← rev * 10 + d
n ← n DIV 10
END WHILE
PRINT rev
```

A. 1009

B. 9001

C. 1010

D. 1